

Besher Alkurdi

NLP Research Engineer

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Education

University of Jyväskylä <i>MSc. in Artificial Intelligence</i>	Ongoing
Yildiz Technical University <i>B.S. in Computer Science, 2nd in class</i>	June 2023 GPA: 3.84/4.0

Experience

Research Engineer at Huawei	Aug 2023 - Aug 2024
- Curated and synthesized data for fine-tuning in multiple languages specific to customer requirements. - Used custom model and data parallelism with mixed precision for large language model finetuning. - Enhanced the finetuning processes by implementing parameter-efficient training, advanced context expansion, retrieval-augmented generation, and sampling techniques. - Created and maintained a versatile model serving platform for easy access to fine-tuned models via RESTful and streaming APIs. - Implemented LLM architectures to run on different hardware and be compatible with different ML libraries. - Skills: Natural Language Processing (NLP), MLOps, PyTorch, DeepSpeed	
Undergraduate Research Fellow at Yildiz Technical University	June 2022 - March 2023
- Worked under the "Sort Optimization of Training Samples in Machine Learning" project, publicly funded. - Investigated and implemented curriculum learning methods. - Conducted research on data augmentation using several methods. - Analyzed the impact of distributed training on the training process. - Experimented with loss function modifications for deep neural networks. - Trained multimodal models using contrastive learning. - Skills: Natural Language Processing (NLP), Python, TensorFlow, PyTorch	
Research Intern at New Mind Bilgi Yönetim Sistemleri	July 2021 - October 2021
- Conducted research on state-of-the-art natural language processing techniques and their application in a system assisting lawyers in lawsuits. - Focused on Document AI applications in the Turkish language. - Skills: Research, Python, Keras	

Publications

Scaling Up Paraphrase Generation Datasets with MT and Semantic Similarity Filtering	2024
An extension of our prior work. Presented a methodology for filtering sentence pairs in parallel corpora, using machine translation and semantic similarity. Experimented with data augmentation using paraphrasing.	
Semantic Similarity Based Filtering for Turkish Paraphrase Dataset Creation	December 2022
We introduced a method for creating paraphrase datasets from parallel bilingual corpora along the largest paraphrase datasets in Turkish.	
Linguistic-based Data Augmentation Approach for Offensive Language Detection	2022
The aim of the study is to augment classification datasets using a linguistic-based data augmentation approach to improve the detection of offensive language and reduce social bias.	

Certificates

Deep Reinforcement Learning Course <i>HuggingFace</i> Reinforcement Learning and Deep Reinforcement Learning	May 2024
Generative Adversarial Networks (GANs) Specialization <i>DeepLearning.AI</i> GAN components, data augmentation, privacy.	July 2023
Natural Language Processing Specialization <i>DeepLearning.AI</i> NLP, Transformers, Attention models, Machine Translation.	June 2023
Deep Learning Specialization <i>DeepLearning.AI</i> Deep learning, including GANs and VAEs.	February 2023
Machine Learning Specialization <i>DeepLearning.AI</i> ML fundamentals and practical applications.	August 2022
Deep Learning in Python Track <i>DataCamp</i> Keras, PyTorch, deep learning workflows, and model optimization.	May 2022
Introduction to Computational Thinking and Data Science <i>edX</i> Introductory course in data science and ML.	May 2021

Verification: All certificates are verifiable at <https://www.linkedin.com/in/mrbesher>.

Volunteer Experience

Schoolhouse.world, Tutor Tutoring students worldwide in subjects such as Math, Computer Science, and more.	February 28, 2021 - Present
Al-Aqsa Youth Institution, Tutor Taught computer, networking, and programming basics to middle school and high school refugee students.	November 30, 2020 - August 31, 2021
Ask Me, Translator Assisted tourists visiting Istanbul by providing information about the city's history and attractions as part of Fatih Municipality's "Voluntary Tourism Delegates" program.	June 30, 2019 - August 31, 2019

Projects

Nano BERT and decoder | *GitHub:* github.com/mrbesher/nanobert, github.com/mrbesher/nanodecoder

A reimplementation of BERT with ALiBi trained on characters as tokens, and an ongoing work to implement a GPT architecture with enhancements like SwiGLU, RoPE, Pre-LN, RMSNorm, multi-head latent attention, and sparse mixture of experts to improve activation, positional encoding, normalization, inference efficiency, and dynamic input processing.

Gaze Tracking Robot | *GitHub:* github.com/mrbesher/gaze-tracking-robot

Developed a Gaze Controlled Robot designed to empower individuals with disabilities to control wheelchairs using eye movements. Used ESP32, OpenCV, MediaPipe, and Tkinter.

The Genetic Art Forger | *GitHub:* github.com/mrbesher/genetic-art-forger

An experiment with genetic algorithms to generate art. The project aimed to create an image starting from the bottom-left corner, without skipping any pixels (analogous to not lifting the "pencil").

Languages

Arabic: Native • **English:** Advanced • **Turkish:** Advanced • **Finnish:** Beginner • **French:** Beginner

Skills

Tools and Tech: Linux, Git/GitHub, PyTorch, TensorFlow, Keras, OpenCV, ROS, TRAX, NLP

Coding Languages: Python, C, Rust, SQL, Java